

Instructor Key: Day 4 Mastery Quiz

Comparisons and Boundaries

Key for the Day 4 student quiz package.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 4 mastery quiz; quiz, not workbooklet.
Content anchor	Built after Lance approved the Day 4 workbooklet content surface: comparison operators, inclusive/exclusive boundaries, equality, and type-safe boundary checks.
Difficulty posture	Higher than Day 3 because it combines exact-limit operator choice, equality-versus-assignment interpretation, a TypeError-style type-transfer trap, and a no-and-yet preparation item.
Scenario/source boundary	Lower-sensitivity pump-station threshold record. No protected exam/review prompt text, answer-key language, private student data, or near-copy source structure used.
Student-surface rule	Student PDF intentionally hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.
Copyright	Copyright © 2026 Lance L.A. White. All rights reserved.

2 Reference code

```
1 # converted values from the Day 3 intake process
2 target_lpm = 21.0
3 tolerance_lpm = 0.6
4 low_limit_lpm = target_lpm - tolerance_lpm
5 high_limit_lpm = target_lpm + tolerance_lpm
6 reading_low_edge = 20.4
7 reading_mid = 21.0
8 reading_high_edge = 21.6
9 reading_over = 21.61
10 reading_text = "21.6"
11
12 # individual comparison checks
13 reading_low_edge >= low_limit_lpm
14 reading_low_edge > low_limit_lpm
15 reading_high_edge <= high_limit_lpm
16 reading_high_edge < high_limit_lpm
17 reading_over <= high_limit_lpm
18 reading_mid == target_lpm
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Common issue / support move	Tags
1	Limits from converted values	<code>low_limit_lpm = 20.4;</code> <code>high_limit_lpm = 21.6.</code>	If student uses text values, return to Day 3 cast evidence.	trace; predict
1	Lower inclusive boundary	True. <code>20.4 >= 20.4</code> includes equality at the lower boundary.	If false, ask what the equals bar in <code>>=</code> means.	predict; explain
1	Lower exclusive boundary	False. <code>20.4 > 20.4</code> does not include equality.	Exact-boundary trap: "greater than" means strictly above.	predict; explain
1	Upper inclusive boundary	True. <code>21.6 <= 21.6</code> includes equality at the upper boundary.	If false, compare <code><=</code> with <code><</code> .	predict; explain
1	Upper exclusive boundary	False. <code>21.6 < 21.6</code> does not include equality.	Ask student to test just below and exactly at the boundary.	predict; explain
1	Above-boundary miss	False. <code>21.61 <= 21.6</code> fails because the reading is above the upper limit.	If true, have student order <code>21.61</code> and <code>21.6</code> on a number line.	test; debug
1	Equality check, not assignment	True. <code>reading_mid == target_lpm</code> checks equality; it does not assign or overwrite a value.	If student describes storage change, contrast <code>=</code> and <code>==</code> .	recognize; explain
1	Type-transfer boundary trap	Not a safe Python comparison: <code>reading_text</code> is a string and <code>high_limit_lpm</code> is a float. Convert first, e.g., <code>float(reading_text) <= high_limit_lpm</code> .	If student answers only true/false, ask what types are being compared.	debug; test
1	Boundary-note rewrite	Secure rewrite: both exact-limit readings pass under inclusive rules: <code>20.4 >= 20.4</code> and <code>21.6 <= 21.6</code> . They would fail only under exclusive <code>></code> or <code><</code> rules.	Require inclusive/exclusive language for both edges.	debug; explain
1	Prepare Day 5 evidence	Separate checks for <code>reading_high_edge</code> : <code>reading_high_edge >= low_limit_lpm</code> and <code>reading_high_edge <= high_limit_lpm</code> . Do not connect with and yet.	If student combines with and , the logic may be right but violates the Day 4 separate-evidence constraint.	implement; transfer

4 Expected comparison trace

Expression	Result	Boundary reason
<code>reading_low_edge >= low_limit_lpm</code>	True	Equality is included by <code>>=</code> .
<code>reading_low_edge > low_limit_lpm</code>	False	Equality is excluded by <code>></code> .
<code>reading_high_edge <= high_limit_lpm</code>	True	Equality is included by <code><=</code> .
<code>reading_high_edge < high_limit_lpm</code>	False	Equality is excluded by <code><</code> .
<code>reading_over <= high_limit_lpm</code>	False	<code>21.61</code> is above <code>21.6</code> .
<code>reading_mid == target_lpm</code>	True	Both numeric values are <code>21.0</code> .

5 Mastery interpretation and support routing

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10, including exact-boundary reasoning	Student can distinguish inclusive/exclusive operators, equality checks, and type-safe numeric comparisons.	Continue to Day 5 Boolean combinations using separate lower/upper evidence.
Developing	5–7 / 10, or one boundary edge reversed	Student can compute limits but may miss the equality bar or forget Day 3 type/cast evidence.	Target exact lower/upper edge rows and one text-versus-float comparison example.
Needs support	0–4 / 10	Student does not yet connect operator symbols to exact-boundary outcomes.	Use just-below, exact, and just-above rows for one limit before returning to two-sided ranges.

6 Optional recovery mini-surface

Use if several students miss inclusive/exclusive boundary meaning.

```
limit = 10
value = 10
value >= limit # True: equality included
value > limit  # False: equality excluded
value <= limit # True: equality included
value < limit  # False: equality excluded
```

Secure recovery answer: the equals bar decides whether exact boundary values are included.

7 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, print-safe margins, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.